UPCISS

O-Level (M4-R5)

Internet of Things and its Applications

Chapter-4

Interfacing Different Types of Sensor and Actuators

Video Tutorial

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Contents

Interfacing Different Types of Sensor and Actuators ---------- 2

Push Button.....2

Buzzer (Piezo).....2

Touch Sensor.....3

RGB LED Light.....4

Ultrasonic Sensor (HC-SR04).....5

LDR Sensor.....5

7 Segment LED Display.....6

MQ135 Gas Sensor.....9

PIR Motion Sensor (Passive Infrared Sensor).....9

DHT 11 Humidity and Temperature Sensor.....10

Interfacing Keypad 4X4.....10

Interfacing LCD 16X2.....11

DHT11 Sensor Print on LCD.....12

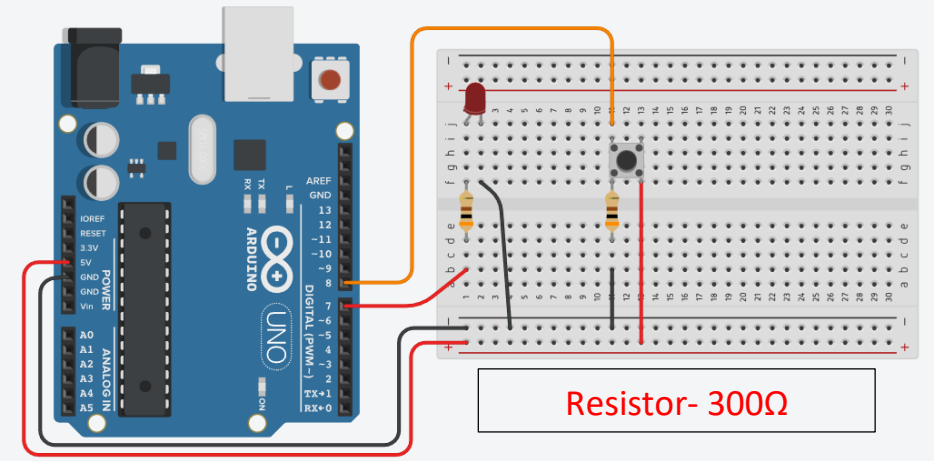
IR (Infrared) Sensor.....12

HC-05 Bluetooth Module.....13

Servo Motor.....13

Interfacing Different Types of Sensor and Actuators

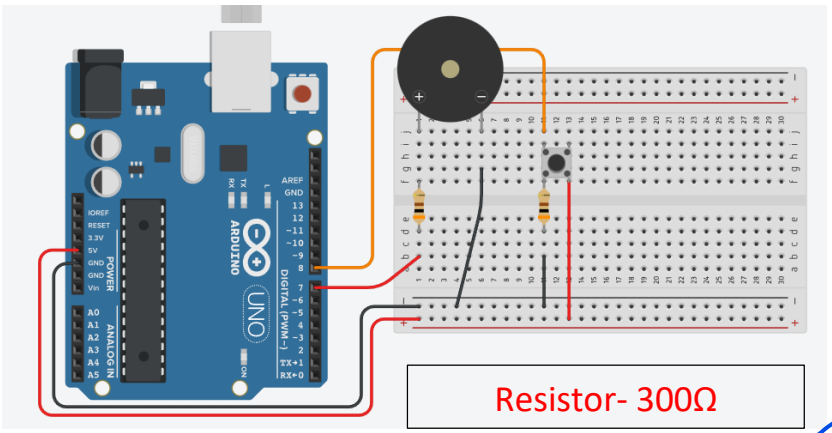
Push Button




```
int led = 7, pb = 8;
void setup() {
  Serial.begin(9600);
  pinMode(led, OUTPUT);
  pinMode(pb, INPUT);
}
void loop() {
  int signal = digitalRead(pb);
  Serial.print (signal);
  if (signal == 1) {
    digitalWrite(led, 1);
    delay(200);
  } else {
    digitalWrite(led, 0);
  }
}
```

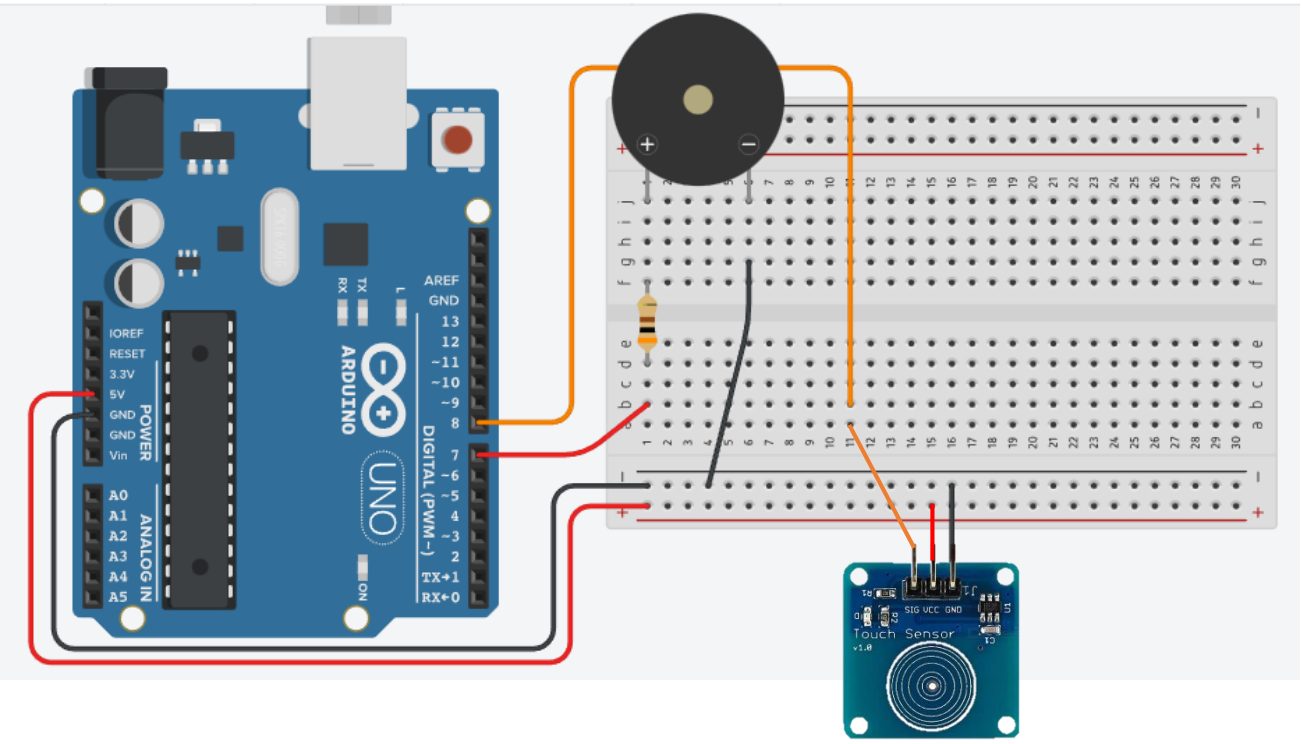
Same

Buzzer (Piezo)




```
int buz = 7, pb = 8;
void setup() {
  Serial.begin(9600);
  pinMode(buz, OUTPUT);
  pinMode(pb, INPUT);
}
void loop() {
  int signal = digitalRead(pb);
  Serial.print (signal);
  if (signal == 1) {
    digitalWrite(buz, 1);
    delay(200);
  } else {
    digitalWrite(buz, 0);
  }
}
```

Touch Sensor




```
int buz = 7, tch = 8;
void setup() {
  Serial.begin(9600);
  pinMode(buz, OUTPUT);
  pinMode(tch, INPUT);
}
void loop() {
  int signal = digitalRead(tch);
  Serial.print (signal);
  if (signal == 1) {
    digitalWrite(buz, 1);
    delay(200);
  } else {
    digitalWrite(buz, 0);
  }
}
```

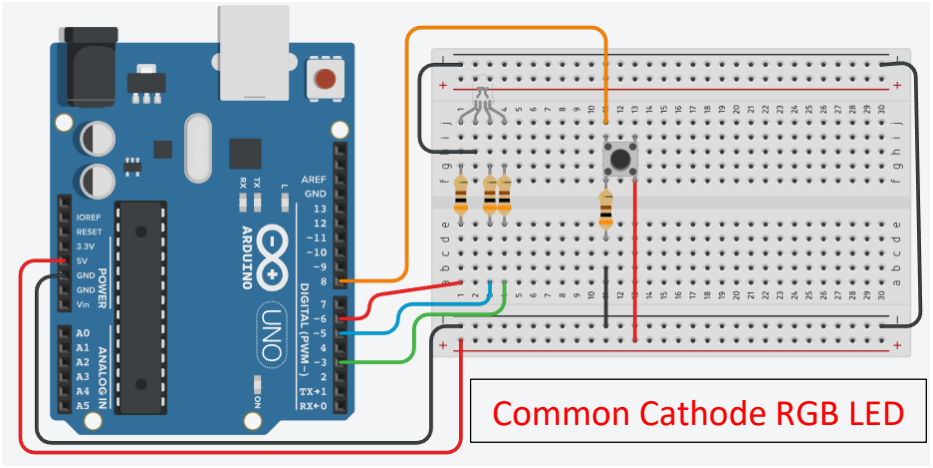
Same ✓

RGB LED Light

```
int red = 6, blue = 5, green = 3, pb = 8, c = 0;
void setup() {
  pinMode(red, OUTPUT);
  pinMode(blue, OUTPUT);
  pinMode(green, OUTPUT);
  pinMode(pb, INPUT);
}

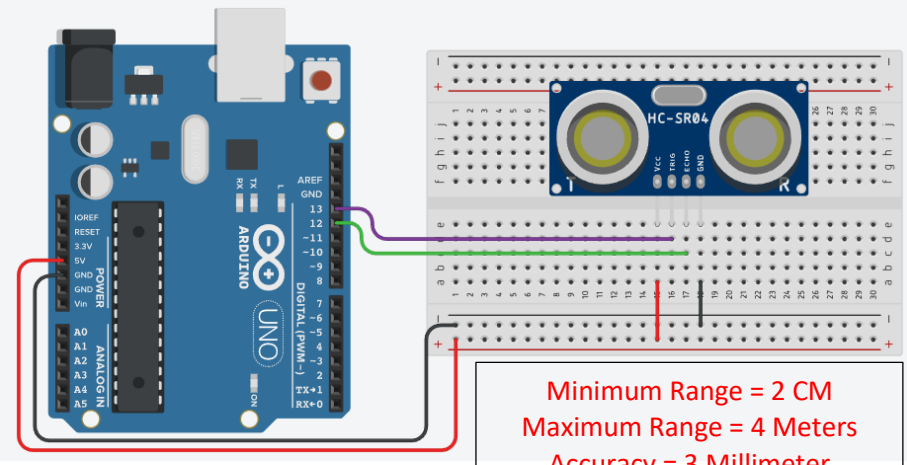
void loop() {
  int signal = digitalRead(pb);
  if (signal == 1) { c++; }
  if (c == 1) {
    setcolor(255, 0, 0);
    delay(100);
    setcolor(160, 110, 90);
    delay(100);
    setcolor(200, 70, 190);
    delay(100);
    setcolor(255, 170, 80);
    delay(100);
  } else if (c == 2) {
    setcolor(0, 255, 0);
    delay(100);
    setcolor(60, 210, 50);
    delay(100);
    setcolor(100, 240, 90);
    delay(100);
    setcolor(110, 255, 50);
    delay(100);
  } else {
    setcolor(0, 0, 0);
    delay(200);
    c = 0;
  }
}

void setcolor(int r, int b, int g) {
  analogWrite(red, r);
  analogWrite(blue, b);
  analogWrite(green, g);
}
```



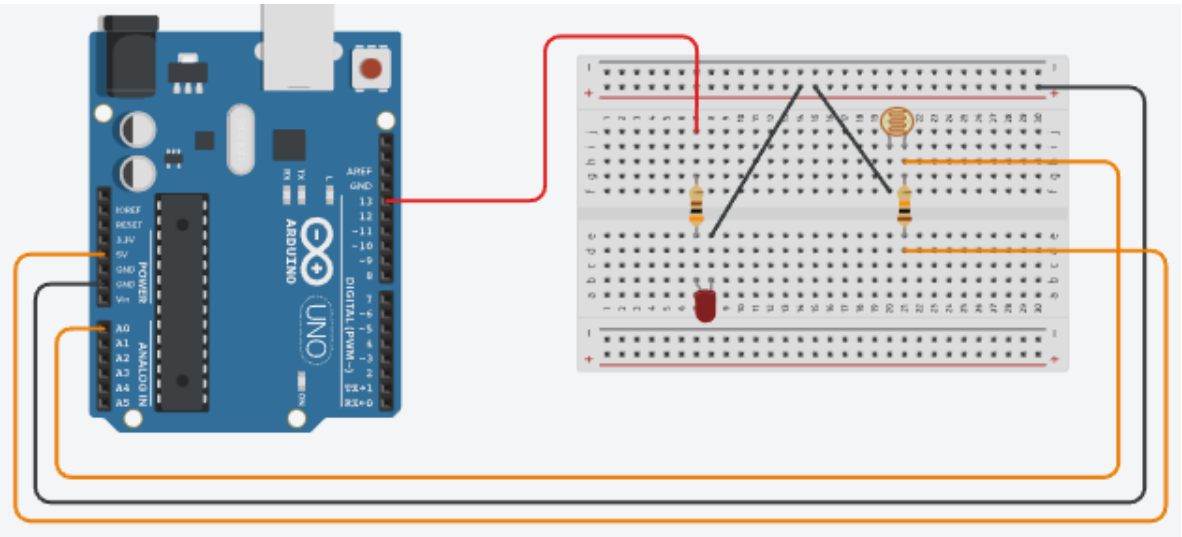
Ultrasonic Sensor (HC-SR04)

```
int trig = 13;
int echo = 12;
int duration;
int distance;
void setup() {
  Serial.begin(9600);
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
}
void loop() {
  digitalWrite(trig, 0);
  delayMicroseconds(2);
  digitalWrite(trig, 1);
  delayMicroseconds(10);
  digitalWrite(trig, 0);
  duration = pulseIn(echo, 1);  // in Microseconds
  distance = duration * 0.034 / 2;
  Serial.print("Distance in CM: ");
  Serial.println(distance);
}
```



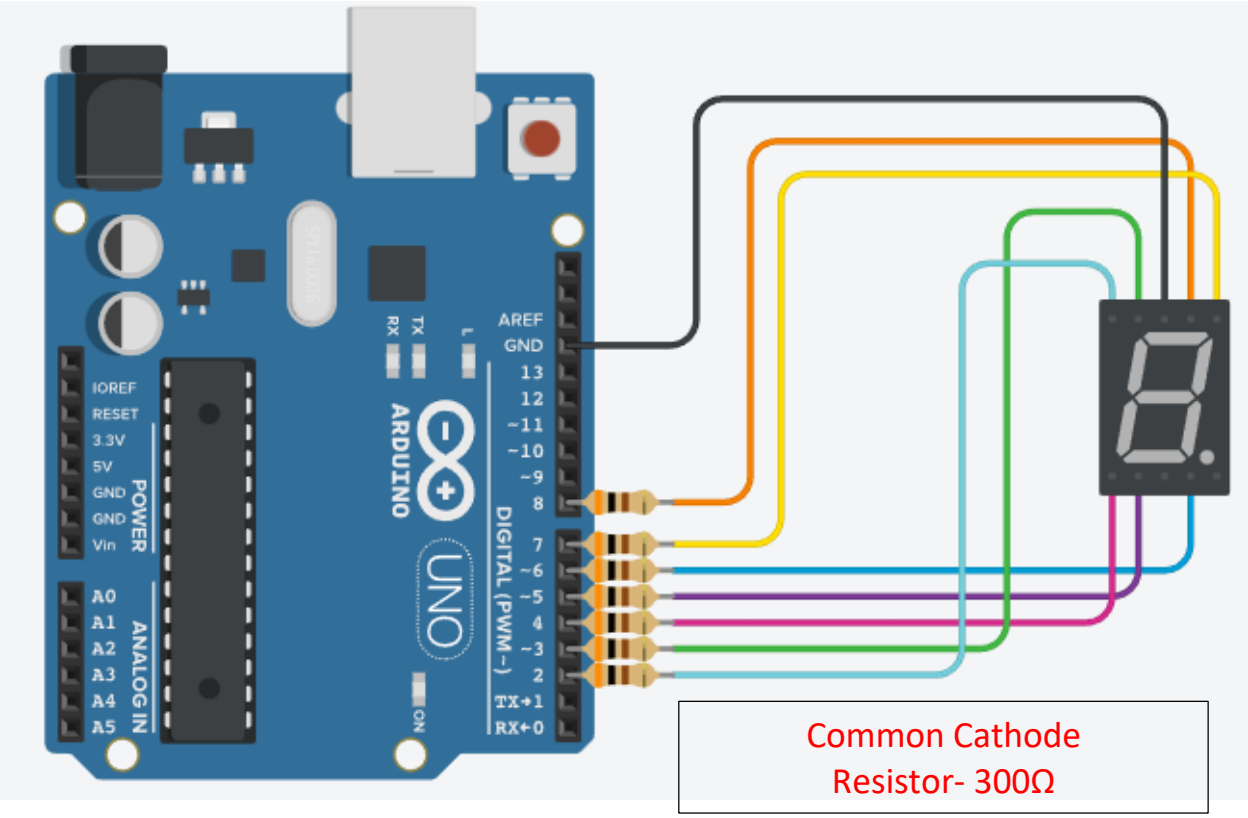
Minimum Range = 2 CM
Maximum Range = 4 Meters
Accuracy = 3 Millimeter

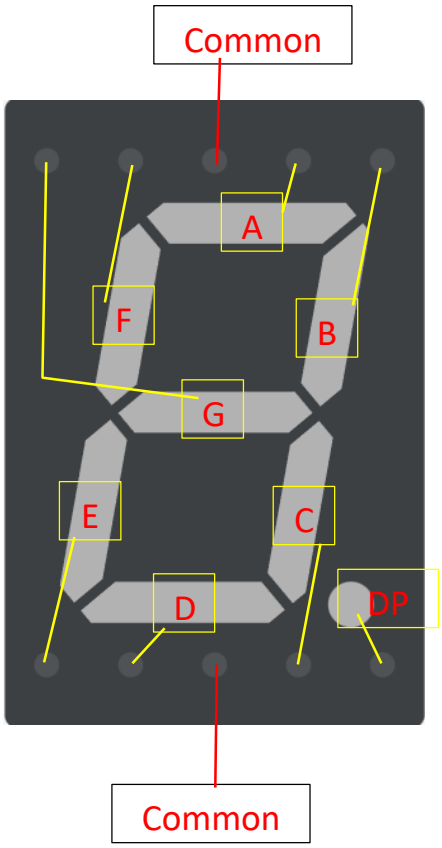
LDR Sensor



```
void setup() {
  pinMode(13, OUTPUT);
  Serial.begin(9600);
}
void loop() {
  int ldr = analogRead(A0);
  Serial.println(ldr);
  delay(100);
  if (ldr > 900) { digitalWrite(13, 1);}
  else {digitalWrite(13, 0);}
}
```

7 Segment LED Display

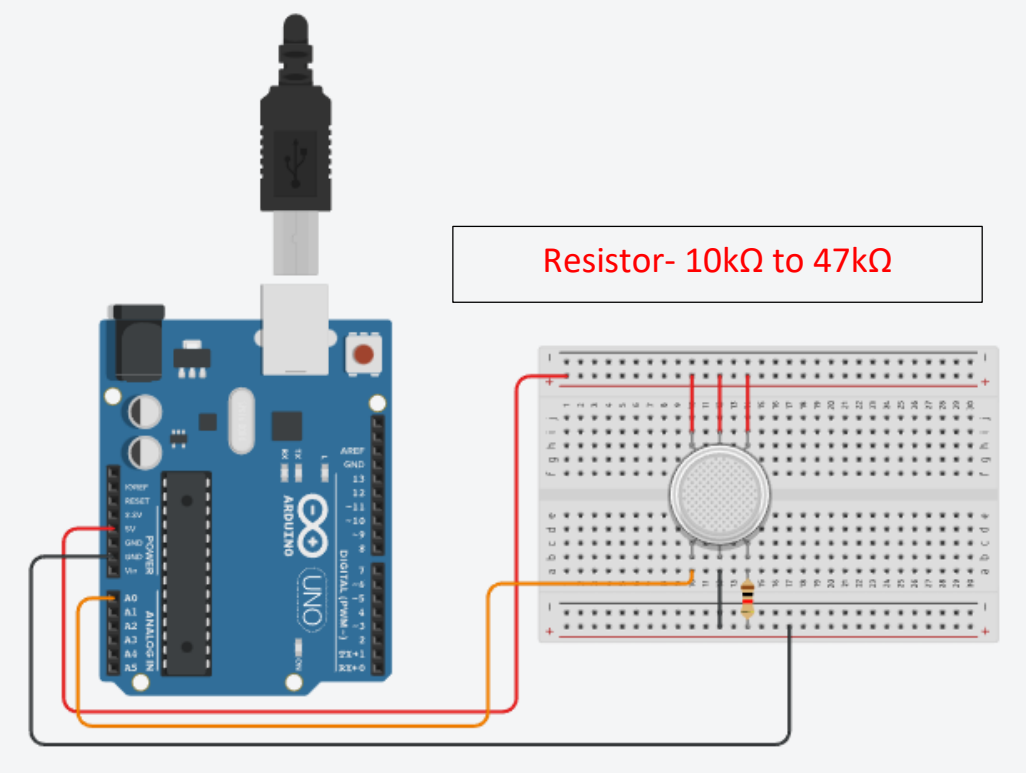




```
int a = 8, b = 7, c = 6, d = 5, e = 4, f = 3, g = 2;
int dt = 1000;
void setup() {
  pinMode(a, OUTPUT);
  pinMode(b, OUTPUT);
  pinMode(c, OUTPUT);
  pinMode(d, OUTPUT);
  pinMode(e, OUTPUT);
  pinMode(f, OUTPUT);
  pinMode(g, OUTPUT);
}
void loop() {
  // Write 0
  digitalWrite(a, 1);
  digitalWrite(b, 1);
  digitalWrite(c, 1);
  digitalWrite(d, 1);
  digitalWrite(e, 1);
  digitalWrite(f, 1);
  digitalWrite(g, 0);
  delay(dt);
  // Write 1
  digitalWrite(a, 0);
  digitalWrite(b, 1);
  digitalWrite(c, 1);
  digitalWrite(d, 0);
  digitalWrite(e, 0);
  digitalWrite(f, 0);
  digitalWrite(g, 0);
  delay(dt);
  // Write 2
  digitalWrite(a, 1);
  digitalWrite(b, 1);
  digitalWrite(c, 0);
  digitalWrite(d, 1);
  digitalWrite(e, 1);
  digitalWrite(f, 0);
  digitalWrite(g, 1);
  delay(dt);
  // Write 3
  digitalWrite(a, 1);
  digitalWrite(b, 1);
  digitalWrite(c, 1);
  digitalWrite(d, 1);
  digitalWrite(e, 0);
  digitalWrite(f, 0);
  digitalWrite(g, 1);
  delay(dt);
  // Write 4
  digitalWrite(a, 0);
  digitalWrite(b, 1);
  digitalWrite(c, 1);
```

```
  digitalWrite(d, 0);
  digitalWrite(e, 0);
  digitalWrite(f, 1);
  digitalWrite(g, 1);
  delay(dt);
  // Write 5
  digitalWrite(a, 1);
  digitalWrite(b, 0);
  digitalWrite(c, 1);
  digitalWrite(d, 1);
  digitalWrite(e, 0);
  digitalWrite(f, 1);
  digitalWrite(g, 1);
  delay(dt);
  // Write 6
  digitalWrite(a, 1);
  digitalWrite(b, 1);
  digitalWrite(c, 1);
  digitalWrite(d, 1);
  digitalWrite(e, 1);
  digitalWrite(f, 0);
  digitalWrite(g, 1);
  delay(dt);
  // Write 7
  digitalWrite(a, 1);
  digitalWrite(b, 1);
  digitalWrite(c, 1);
  digitalWrite(d, 0);
  digitalWrite(e, 0);
  digitalWrite(f, 0);
  digitalWrite(g, 0);
  delay(dt);
  // Write 8
  digitalWrite(a, 1);
  digitalWrite(b, 1);
  digitalWrite(c, 1);
  digitalWrite(d, 1);
  digitalWrite(e, 1);
  digitalWrite(f, 1);
  digitalWrite(g, 1);
  delay(dt);
  // Write 9
  digitalWrite(a, 1);
  digitalWrite(b, 1);
  digitalWrite(c, 1);
  digitalWrite(d, 1);
  digitalWrite(e, 0);
  digitalWrite(f, 1);
  digitalWrite(g, 1);
  delay(dt);
}
```

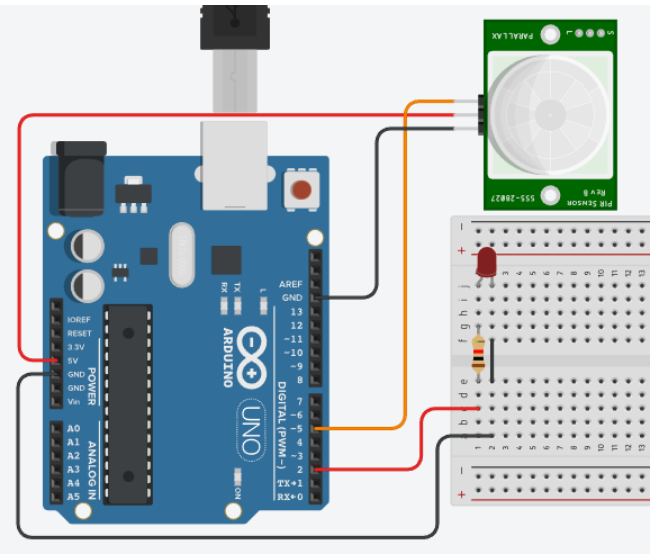
MQ135 Gas Sensor




```
void setup() {
  Serial.begin(9600);
}
void loop() {
  int val = analogRead(A0);
  int air = map(val, 0, 1023, 0, 255);
  Serial.print("Air Quality ");
  Serial.println(air);
  delay(1000);
}
```

PIR Motion Sensor (Passive Infrared Sensor)

```
void setup() {
  pinMode(2, OUTPUT);
  pinMode(5, INPUT);
  Serial.begin(9600);
}
void loop() {
  int value = digitalRead(5);
  Serial.println(value);
  if (value == 1) {
    digitalWrite(2, 1);
  } else {
    digitalWrite(2, 0);
  }
}
```



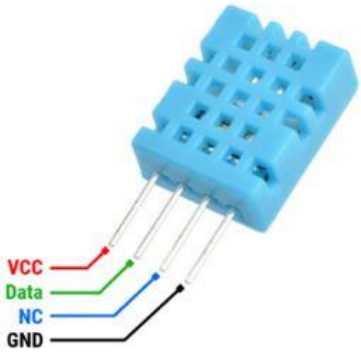
DHT 11 Humidity and Temperature Sensor

```
#include <DHT.h>
#include <DHT_U.h>

DHT dht(2, DHT11);

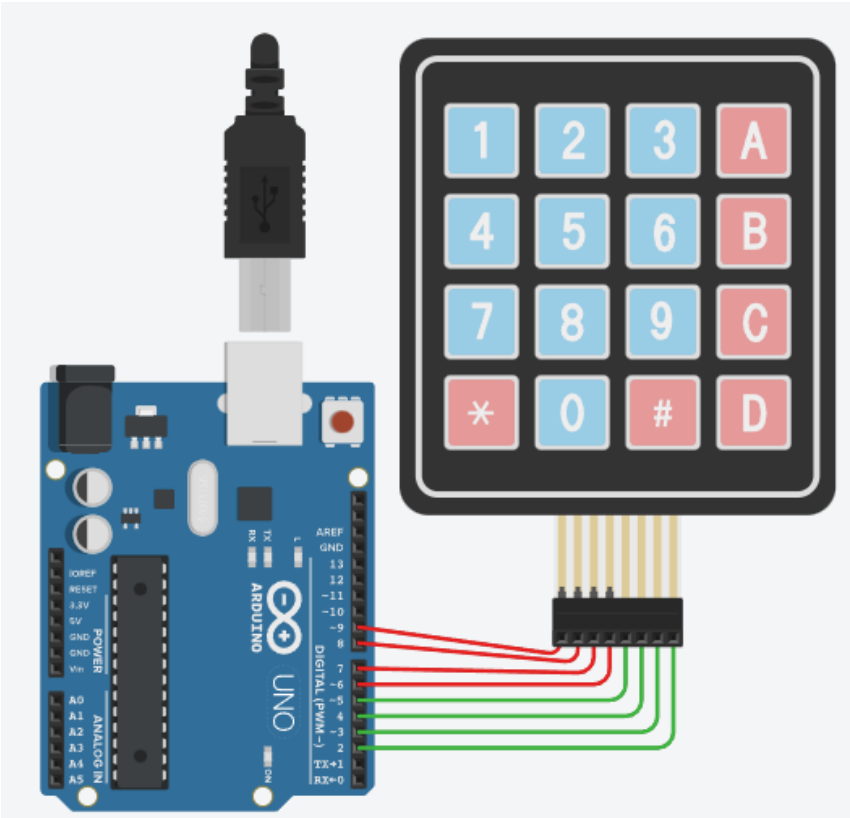
void setup() {
  Serial.begin(9600);
  dht.begin();
}
void loop() {
  delay(2000);
  float h = dht.readHumidity();
  float tc = dht.readTemperature();
  float tf = dht.readTemperature(true);

  Serial.print(h);
  Serial.print(" % Humidity \t");
  Serial.print(tc);
  Serial.print(" °C Temperature \t");
  Serial.print(tf);
  Serial.print(" °F Temperature \t");
  Serial.println("");
}
```



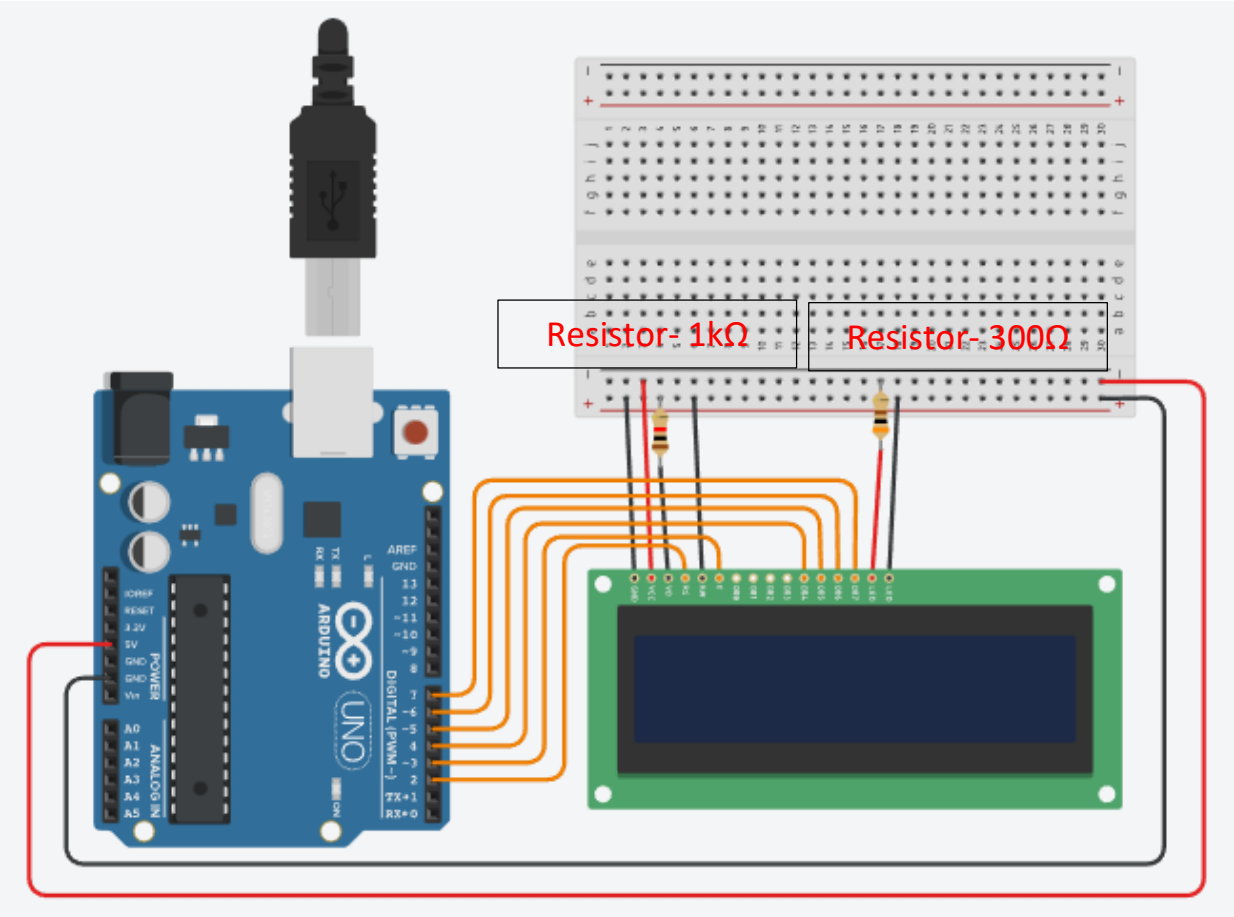
- Humidity Range: 20-90% RH
- Humidity Accuracy: ±5% RH
- Temperature Range: 0-50 °C
- Temperature Accuracy: ±2% °C
- Operating Voltage: 3V to 5.5V

Interfacing Keypad 4X4



```
#include <Keypad.h>
byte row = 4;
byte col = 4;
char keys[4][4] = {
  { '1', '2', '3', 'A' },
  { '4', '5', '6', 'B' },
  { '7', '8', '9', 'C' },
  { '*', '0', '#', 'D' }
};
byte rowpin[4] = { 9, 8, 7, 6 };
byte colpin[4] = { 5, 4, 3, 2 };
Keypad mykeypad = Keypad(makeKeymap(keys), rowpin, colpin, row, col);
void setup() {
  Serial.begin(9600);
}
void loop() {
  char presskey = mykeypad.getKey();
  if (presskey) {
    Serial.println(presskey);
  }
}
```

Interfacing LCD 16X2



```
#include <LiquidCrystal.h>
LiquidCrystal lcd(2, 3, 4, 5, 6, 7);
void setup() {
  lcd.begin(16, 2);
  lcd.print("Hello UPCISS");
}
void loop() {
  lcd.setCursor(0, 1);
  lcd.print(millis() / 1000);
}
```

DHT11 Sensor Print on LCD

```
#include <DHT.h>
#include <DHT_U.h>
#include <LiquidCrystal.h>
DHT dht(8, DHT11);
LiquidCrystal lcd(2, 3, 4, 5, 6, 7);
void setup() {
  Serial.begin(9600);
  dht.begin();
  lcd.begin(16, 2);
}
void loop() {
  float h = dht.readHumidity();
  float tc = dht.readTemperature();
  lcd.print(h);
  lcd.print(" % Humidity");
  lcd.setCursor(0, 1);
  lcd.print(tc);
  lcd.print(" °C Temperature");
  delay(2000);
  lcd.clear();
}
```

IR (Infrared) Sensor

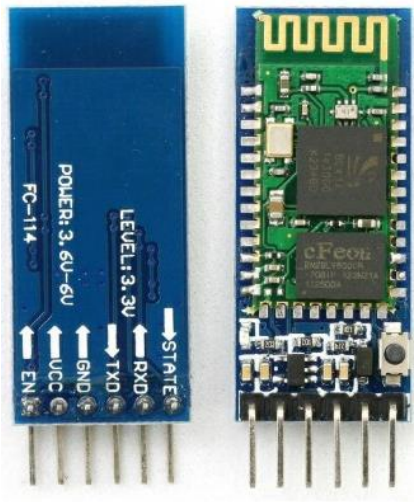





```
#include <IRremote.h>
int irs = 5;
int c = 1;
void setup() {
  Serial.begin(9600);
  pinMode(2, OUTPUT);
  IrReceiver.begin(irs, ENABLE_LED_FEEDBACK);
}
void loop() {
  if (IrReceiver.decode()) {
    String ircode = String(IrReceiver.decodedIRData.command, HEX);
    delay(300);
    Serial.println(ircode);
    IrReceiver.resume();
```

```
    if ((ircode == "f") && (c % 2 == 1)) {
      digitalWrite(2, 1);
      c++;
    } else if ((ircode == "f") && (c % 2 == 0)) {
      digitalWrite(2, 0);
      c++;
    }
  }
}
```

HC-05 Bluetooth Module




```
char val;
void setup() {
  Serial.begin(9600);
  pinMode(2, OUTPUT);
}
void loop() {
  if (Serial.available()) {
    val = Serial.read();
    if (val == 'A') {
      digitalWrite(2, 1);
    } else if (val == 'a') {
      digitalWrite(2, 0);
    }
  }
}
```

Servo Motor




```
#include <Servo.h>
Servo myservo;
void setup() {
  myservo.attach(7);
}
void loop() {
  for (int i = 0; i <= 180; i++) {
    myservo.write(i);
    delay(10);
  }
  for (int i = 180; i > 0; i--) {
    myservo.write(i);
    delay(10);
  }
}
```

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